Generative AI Agents for Real-Time Player Support in Video Games

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Generative AI is increasingly being adopted across a wide range of domains. Tools such as chatbots and other LLM-powered applications are being developed for virtually every task, offering users support in many different contexts. However, unlike the traditional interaction paradigm of most LLM-based tools—where the user must explicitly provide input—Generative AI–powered agents are designed to act proactively and continuously. These agents can operate with minimal direct user input, providing real-time assistance and adapting dynamically to ongoing activities.

One domain where such agents are still largely unexplored is video games. While Microsoft has recently released *Copilot for Games* as a beta, the tool is currently very limited and does not offer meaningful real-time, in-game support. This represents a missed opportunity: a well-designed AI agent could enhance player enjoyment, improve performance, and reduce player churn, contributing to the sustainability of video game ecosystems.

This thesis proposal takes a concrete and implementation-oriented approach. Through a Design Science Research process and careful engineering, the goal is to design and develop a prototype of a **local Generative AI agent** capable of interacting with a player's game in real time. The agent may provide different forms of support, potentially also leveraging connections to external services or activities. Students are encouraged to explore and refine their ideas autonomously, in close discussion with the supervisor. The thesis will conclude with a small-scale validation involving end users to assess the usefulness and feasibility of the proposed solution.

A potential process could involve the following steps:

1. Review of the current state of the art (some references are provided below)
2. Interviews and research with players to understand their needs and issues

3. Development of a prototype (including prompt engineering techniques) and evaluation of the prototype with players in order to test (1) technologies and (2) the prototype’s potential impact on the gaming experience.

Potential collaborations with companies in Denmark can be arranged. The professor already has some contacts with game startups in Denmark, but the plan is to search for collaborations with others.

References:

https://consensus.app/search/ai-agents-for-video-gaming/tgZTLIs2Qq6kKSs3DpOJ8A/?utm_source=share&utm_medium=clipboard